Learning Objective 3.3: I can implement an FM receiver in GNU Radio, demodulate a live broadcast FM signal, and decode its RDS data.

Learning Objective 3.4: I can explain and demonstrate the trade-off between bandwidth and noise performance for AM vs. FM.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **FM bandwidth.** Carson's rule estimates the bandwidth of an FM signal with peak deviation Δf and audio bandwidth W as

$$B \approx 2(\Delta f + W)$$

(a) Fill in the table. For the AM row, the station sends the same audio as DSB-AM.

Service	Δf (kHz)	W (kHz)	Bandwidth (kHz)
Broadcast FM, mono	75	15	
Two-way radio, NBFM	5	3	
AM, broadcast audio	–	15	

Broadcast FM: $2(75+15) = 180$ kHz. NBFM: $2(5+3) = 16$ kHz. AM: $2W = 30$ kHz. Broadcast FM is six times as wide as AM carrying the same audio.

(b) Open the FM Explorer on the Lesson 7 page and choose the broadcast preset, 75 kHz of deviation with a 15 kHz tone. Read the power inside Carson's band, and explain in words why the signal is wider than $2\Delta f$ = 150 kHz.

Carson's band, 180 kHz, holds 99.4% of the power. The explorer draws lines every 15 kHz out past ± 75 kHz: the frequency swings at the tone's rate, and that swing adds lines beyond the peak deviation, about one tone frequency on each side.

2. **The WBFM receiver.** An RTL-SDR records 2.4 MS/s centered on 98.8 MHz, and the station we want is at 99.1 MHz. The receiver is `L07-wbfm-rx.grc`.

   (a) Find the center frequency of the Frequency Xlating FIR Filter, and the sample rate after it decimates by 10.

   $$99.1 - 98.8 = 0.3 \text{ MHz} = 300 \text{ kHz}$$

   $$\frac{2.4 \text{ MS/s}}{10} = 240 \text{ kS/s}$$

   shift = 300 kHz

   rate = 240 kS/s

   (b) A classmate decimates by 20 instead. Explain what goes wrong, using the channel filter's ±90 kHz passband and the 57 kHz RDS subcarrier.

   At 120 kS/s the flowgraph can hold only -60 to $+60$ kHz. The station's ±90 kHz no longer fits, so its outer parts alias into the band, and after demodulation the RDS signal, which reaches 59.4 kHz, sits at the edge of the Nyquist frequency with no room for a filter.

   (c) Find the Quadrature Demod gain that makes 75 kHz of deviation produce an output of 1, and the output when the carrier sits 30 kHz above center.

   $$\frac{240 \text{ kHz}}{2\pi \times 75 \text{ kHz}} = 0.509$$

   $$\frac{30}{75} = 0.40$$

   gain = 0.509

   output = 0.40

   (d) Find the decimation from the demodulator to a 48 kHz sound card, and the corner frequency of a 75 μs de-emphasis filter.

   $$\frac{240}{48} = 5$$

   $$f = \frac{1}{2\pi(75 \times 10^{-6})} = 2.12 \text{ kHz}$$

   decimation = 5

   corner = 2.12 kHz

   (e) Explain why the flowgraph needs no Throttle, whether it plays a file or receives from the RTL-SDR.

   The Audio Sink takes exactly 48,000 samples per second, so it paces a File Source. With the RTL-SDR, the radio's clock and the sound card's already set the pace, and a Throttle would add yet another clock and cause dropouts.

3. **RDS.** RDS sends 1187.5 bits per second on a 57 kHz subcarrier, in groups of 104 bits.

   (a) The RDS branch resamples to 19 kHz. Find the samples per bit, and the time one group takes.

   $$\frac{19000}{1187.5} = 16$$

   $$\frac{104}{1187.5} = 87.6 \text{ ms}$$

   16 samples

   87.6 ms

   (b) The station name is eight characters, sent two per group. Radiotext is up to 64 characters, sent four per group. Find the shortest time to receive each, if the station sent nothing else.

   $$4 \times 87.6 = 350 \text{ ms}$$

   $$16 \times 87.6 = 1.40 \text{ s}$$

   name = 350 ms

   text = 1.40 s

   A station interleaves other group types, so a real receiver takes longer.

   (c) The RDS signal is BPSK, a DSB-SC signal. Name the block in `L07-rds-rx.grc` that recovers its carrier, and explain why the differential coding means the recovered carrier may be 180° off.

   The Constellation Receiver locks to the subcarrier's phase. A BPSK loop can lock at either of two phases 180° apart, which inverts every bit. The data is differentially coded, so the Differential Decoder reads only whether each bit changed from the last, and inverting every bit leaves those changes the same.

   (d) A station's audio is clear, but the station name never appears. Explain why RDS needs a stronger signal than the audio does.

   The noise at the FM demodulator's output rises with frequency, so the 57 kHz subcarrier sees much more noise than the audio below 15 kHz, which de-emphasis also cleans up.

4. **AM and FM in the same noise.** `L07-am-vs-fm.grc` sends a 1 kHz tone as AM at $m = 1$ in a 10 kHz channel and as FM in a $2(\Delta f + 5)$ kHz channel, with the same transmitted power and the same noise. It measured these output SNRs in dB:

C/N in 10 kHz (dB)	30	20	15	10
AM	29.5	19.4	14.4	9.2
FM, Δf = 5 kHz	35.5	25.5	20.5	14.2
FM, Δf = 25 kHz	48.5	39.2	26.1	9.3
FM, Δf = 50 kHz	53.2	39.5	17.3	6.4

(a) Find each FM channel's bandwidth, and how many dB more noise it admits than the 10 kHz AM channel.

5 kHz: $2(5+5) = 20$ kHz, $10\log_{10}(2) = 3.0$ dB . 25 kHz: $2(25+5) = 60$ kHz, $10\log_{10}(6) = 7.8$ dB . 50 kHz: $2(50+5) = 110$ kHz, $10\log_{10}(11) = 10.4$ dB .

20, 60, 110 kHz

3.0, 7.8, 10.4 dB

(b) An FM receiver reaches threshold when its C/N in its own bandwidth falls to about 10 dB. Find the C/N in 10 kHz at which each FM signal reaches threshold, and check each against the table.

13.0, 17.8, 20.4 dB

For 5, 25, and 50 kHz of deviation, the threshold C/N values in 10 kHz are, in dB,

$$10 + 3.0 = 13.0, \quad 10 + 7.8 = 17.8, \quad 10 + 10.4 = 20.4$$

The table agrees. The 5 kHz signal keeps a steady 5 to 6 dB lead over AM down to 10 dB. The 25 kHz signal leads by 19 to 20 dB down to 20 dB C/N and has lost about 7 dB of that lead at 15. The 50 kHz signal leads by 24 dB at 30 but only 20 dB at 20, already near its 20.4 dB threshold.

(c) A link to a distant receiver runs at 10 dB C/N in 10 kHz. Use the table to choose AM or one of the three FM deviations, and explain the choice in words.

Choose FM with 5 kHz deviation, which reaches 14.2 dB. At 10 dB C/N it is about 3 dB below its 13 dB threshold, but its narrow channel admits few clicks, so it keeps 5 of its 6 dB lead over AM. The wider deviations would be better on a strong link, but their wider channels admit enough noise to push them below threshold here, where they are no better than AM or worse.

(d) Above threshold, each doubling of the deviation improves FM's output by about 6 dB. Check that rule against the 30 dB column from 5 to 25 kHz.

Going from 5 to 25 kHz is $\log_2 5 = 2.3$ doublings, and the output rises $48.5 - 35.5 = 13.0$ dB, or 5.6 dB per doubling, close to 6 dB. (From 25 to 50 kHz the meter's own distortion floor, near 57 dB, holds the gain to 4.7 dB.)

5. **An AM receiver for ATIS.** An airport broadcasts its ATIS on 119.25 MHz. The RTL-SDR runs at 2.4 MS/s centered on 119.0 MHz, and the receiver is `L07-am-rx.grc`.

   (a) Find the Frequency Xlating FIR Filter's center frequency, and the decimation that takes it straight to a 48 kHz sound card.

   shift = 250 kHz

   decimation = 50

   $$119.25 - 119.0 = 0.25\,\text{MHz} = 250\,\text{kHz}$$

   $$\frac{2.4\,\text{MS/s}}{48\,\text{kS/s}} = 50$$

   (b) Lesson 6 detected the envelope with Abs, a low-pass filter, and a gain of $\pi/2$. Explain why this receiver needs only Complex to Mag.

   After the Frequency Xlating FIR Filter, the station's carrier sits at 0 Hz, and each complex sample is the envelope times a phase factor. Its magnitude is the envelope itself, with no carrier to rectify or average away, so the receiver needs no $\pi/2$ correction.

   (c) Explain why the AM receiver needs an AGC and the FM receiver does not.

   An envelope detector's output is proportional to the received amplitude, which changes with an aircraft's distance, so the AGC holds the audio level steady. The Quadrature Demod measures only how fast the phase turns, so the FM receiver's output does not depend on the signal's strength.

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